Sonning kvadrat ildizi va kub ildizi

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. $\sqrt{81}$ **nechaga teng?**

A) 3 **B)** 9 **C)** 27 **D)** 40,5

2. $\sqrt[3]{343}$ **nechaga teng?**

A) 49 **B)** 21 **C)** 9 **D)** 7

3. **Qaysi son to'la kvadrat?**

A) 50 **B)** 75 **C)** 140 **D)** 121

4. **Qaysi son to'la kub?**

A) 200 **B)** 90 **C)** 125 **D)** 150

5. **Istalgan** n **natural son uchun** $\sqrt{n^2}$ **nimaga teng?**

A) n^2 **B)** $2n$ **C)** n **D)** $n : 2$

6. $6^2 = 36$ **bo'lgani uchun 6 soni 36 ning … deyiladi.**

A) kubi **B)** kub ildizi
C) kvadrat ildizi **D)** kvadrati

7. $4^3 = 64$ **bo'lgani uchun 4 soni 64 ning … deyiladi.**

A) kubi **B)** kvadrati
C) kvadrat ildizi **D)** kub ildizi

8. $\sqrt{2^4 \cdot 7^2}$ **ni toping.**

A) 14 **B)** 56 **C)** 28 **D)** 196

9. $\sqrt[3]{2^6 \cdot 3^3}$ **ni toping.**

A) 6 **B)** 12 **C)** 36 **D)** 24

10. **Rasmda 1 936 ning tub ko'paytuvchilari ikkita bir xil guruhga ajratilgan.** $\sqrt{1936}$ **nechaga teng?**



A) 968 **B)** 88
C) 22 **D)** 44

11. $\sqrt{50}$ **qaysi ikki ketma-ket natural son orasida?**

A) 7 va 8 **B)** 6 va 7 **C)** 25 va 26 **D)** 8 va 9

12. $\sqrt[3]{20}$ **qaysi ikki ketma-ket natural son orasida?**

A) 4 va 5 **B)** 3 va 4 **C)** 2 va 3 **D)** 6 va 7

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13. Kvadrat shaklidagi stol ustining yuzi 3 600 cm². Uning tomoni qancha?

A) 900 cm **B)** 1 800 cm **C)** 60 cm **D)** 600 cm

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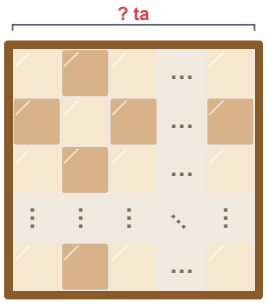
14. Kub shaklidagi idishning hajmi 729 000 cm³. Uning qirrasi qancha?

A) 90 cm **B)** 30 cm

C) 9 cm **D)** 243 000 cm

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15. Kvadrat polga 1 024 ta bir xil plitka terildi (rasm). Polning bir tomonida nechta plitka bor?



A) 32 **B)** 16

C) 64 **D)** 512

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16. Sonning yoyilmasi $2^4 \cdot 3^2 \cdot 5^2$**. U to'la kvadratmi?**

A) Yo'q, darajalar har xil **B)** Yo'q, 5 ning darajasi kichik

C) Ha, hamma darajalar juft **D)** Yo'q, u faqat to'la kub

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17. $3^3 \cdot 7^6$ **to'la kubmi?**

A) Yo'q, 3 ning darajasi toq **B)** Ha, darajalar 3 ga bo'linadi

C) Yo'q, darajalar teng emas **D)** Ha, darajalar juft

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18. Kalkulyatorda sonning kvadrat ildizini topish uchun qaysi tugma bosiladi?

A) $\sqrt{}$ **B)** x^2 **C)** ÷ **D)** %

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19. Qaysi tenglik to'g'ri?

A) $\sqrt[3]{27} = 9$ **B)** $\sqrt{100} = 10$

C) $\sqrt{64} = 4$ **D)** $\sqrt{100} = 50$

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20. Rasmda 1 125 ning tub ko'paytuvchilarini uchta bir xil guruhga ajratishga urinildi. Qaysi xulosa to'g'ri?



A) $\sqrt[3]{1125} = 375$

B) $\sqrt[3]{1125} = 15$

C) 1 125 to'la kub emas

D) $\sqrt[3]{1125} = 5$

Javoblar

Sonning kvadrat ildizi va kub ildizi · 1-test (oson)

№	Javob	Qiymati
1	**B**	9
2	**D**	7
3	**D**	121
4	**C**	125
5	**C**	n
6	**C**	kvadrat ildizi
7	**D**	kub ildizi
8	**C**	28
9	**B**	12
10	**D**	44

№	Javob	Qiymati
11	**A**	7 va 8
12	**C**	2 va 3
13	**C**	60 cm
14	**A**	90 cm
15	**A**	32
16	**C**	Ha, hamma darajalar juft
17	**B**	Ha, darajalar 3 ga bo'linadi
18	**A**	$\sqrt{}$
19	**B**	$\sqrt{100} = 10$
20	**C**	1 125 to'la kub emas

Qisqa yechimlar

1. Qaysi son o'ziga ko'paytirilsa 81 chiqadi? $9 \cdot 9 = 81$.
Demak $\sqrt{81}$ = **9**.
(40,5 — 81 ni 2 ga bo'lish, 27 — 3 ga bo'lish; ildiz bo'lish emas.)

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2. Qaysi son uch marta ko'paytirilsa 343 chiqadi? $7 \cdot 7 \cdot 7 = 343$.
Javob: **7**. ($\sqrt{49} = 7$, lekin 49 — boshqa son.)

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3. $11 \cdot 11 = 121$, $\sqrt{121} = 11$ — natural son.
Javob: **121**. 50, 75, 140 ning kvadrat ildizi natural son emas.

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4. $5 \cdot 5 \cdot 5 = 125$, $\sqrt[3]{125} = 5$.
Javob: **125**. 200 ham to'la kub emas: $5^3 = 125$, $6^3 = 216$.

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5. $n^2 = n \cdot n$ — ikkita bir xil ko'paytuvchi, bittasi n.
Demak $\sqrt{n^2}$ = n. Masalan, $\sqrt{7^2} = 7$.

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6. 36 soni 6 ning kvadrati, 6 soni esa 36 ning **kvadrat ildizi**: $\sqrt{36} = 6$.

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7. 64 soni 4 ning kubi, 4 soni esa 64 ning **kub ildizi**: $\sqrt[3]{64} = 4$.
(64 ning kvadrat ildizi — 8.)

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8. Ikkita bir xil guruh: $2^4 \cdot 7^2 = (2^2 \cdot 7) \cdot (2^2 \cdot 7)$.
Ildiz — bitta guruh: $2^2 \cdot 7 = 4 \cdot 7$ = **28**.
(14 da 2 ning darajasi noto'g'ri kamaytirilgan.)

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9. Uchta bir xil guruh: har birida 2^2 va 3.
Kub ildiz: $2^2 \cdot 3 = 4 \cdot 3$ = **12**.
(24 = $2^3 \cdot 3$ — daraja 3 ga emas, 2 ga bo'lingan.)

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10. Har guruhda: 2, 2, 11.
Ildiz — bitta guruh ko'paytmasi: $2 \cdot 2 \cdot 11$ = **44**.
Tekshirish: $44 \cdot 44 = 1936$. (968 — 2 ga bo'lish xatosi.)

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11. $7 \cdot 7 = 49$, $8 \cdot 8 = 64$, $49 < 50 < 64$.
Demak $\sqrt{50}$ **7 va 8** orasida (kalkulyatorda $\approx 7{,}07$).
(25 va 26 — 50 ni 2 ga bo'lish xatosi.)

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12. $2^3 = 8$, $3^3 = 27$, $8 < 20 < 27$.
Javob: **2 va 3**.
(4 va 5 — $\sqrt{20}$ uchun javob: $16 < 20 < 25$.)

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13. Tomoni = $\sqrt{\text{yuzi}}$. $3600 = 2^4 \cdot 3^2 \cdot 5^2$, $\sqrt{3600} = 2^2 \cdot 3 \cdot 5$ = **60 cm**.
(1 800 — 2 ga bo'lingan, 900 — 4 ga bo'lingan.)

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14. Qirrasi = $\sqrt[3]{\text{hajmi}}$. $729000 = 729 \cdot 1000 = 3^6 \cdot 2^3 \cdot 5^3$.
Kub ildiz: $3^2 \cdot 2 \cdot 5$ = **90 cm**. ($90 \cdot 90 \cdot 90 = 729\,000$.)
9 cm — mingni unutish, 243 000 — 3 ga bo'lish.

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15. Qatorlar soni bir qatordagi plitkalar soniga teng, ya'ni $\sqrt{1024}$ kerak.
$1024 = 2^{10} = 2^5 \cdot 2^5$, $\sqrt{1024} = 2^5$ = **32**.
(512 — 2 ga bo'lish xatosi.)

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16. Darajalar 4, 2, 2 — **hammasi juft**, demak ko'paytuvchilar ikkita bir xil guruhga bo'linadi: to'la kvadrat.
Ildizi: $2^2 \cdot 3 \cdot 5 = 60$.

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17. To'la kubda har bir daraja 3 ga bo'linadi: 3 va 6 — **ha**.
Kub ildizi: $3 \cdot 7^2 = 147$.
(Toq daraja faqat to'la kvadratga xalal beradi.)

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18. Kvadrat ildiz tugmasi — $\sqrt{}$.
x^2 tugmasi aksincha, sonni kvadratga oshiradi.

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19. $10 \cdot 10 = 100$, demak $\sqrt{100} = 10$.
$\sqrt[3]{27} = 3$ (9 emas), $\sqrt{64} = 8$ (4 — bu $\sqrt[3]{64}$).

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20. $1125 = 3^2 \cdot 5^3$: beshliklar guruhlarga bittadan tushdi, ikkita 3 esa ortib qoldi.
Demak **1 125 to'la kub emas** ($10^3 = 1000$, $11^3 = 1331$).
375 — 3 ga bo'lish xatosi.
